

ORF 309 Homework 4 - Question 5 Explanation

Variance of Love Triangles

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Question 5: More Love Triangles

Problem Statement

Recall the dating website from the previous homework:

- There are n people signed up
- For every pair of people, the website independently flips a coin that comes up heads with probability p
- If it comes up heads, the pair is “hooked up”
- A **love triangle** occurs when three people are all mutually hooked up

Previous homework: You computed the expected number of love triangles on the entire site.

This problem: What is the **variance** of the number of love triangles?

Understanding the Setup

Think of this problem as a **random graph**:

- **Vertices (nodes):** The n people
- **Edges:** The hook-ups between pairs of people
- Each possible edge exists independently with probability p
- **Triangle:** Three people forming a complete triangle (all 3 edges present)

There are $\binom{n}{2} = \frac{n(n-1)}{2}$ possible pairs of people (possible edges).

There are $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$ possible triples of people (possible triangles).

Approach Using Indicator Variables

Let T denote the total number of love triangles. We use indicator variables to count them.

For each possible triple of people $\{i, j, k\}$ where $1 \leq i < j < k \leq n$, define:

$$X_{ijk} = \begin{cases} 1 & \text{if people } i, j, k \text{ form a love triangle} \\ 0 & \text{otherwise} \end{cases}$$

Then the total number of triangles is:

$$T = \sum_{1 \leq i < j < k \leq n} X_{ijk}$$

Review: Expected Number of Triangles (Previous Homework)

By linearity of expectation:

$$\begin{aligned} E[T] &= E \left[\sum_{1 \leq i < j < k \leq n} X_{ijk} \right] \\ &= \sum_{1 \leq i < j < k \leq n} E[X_{ijk}] \\ &= \sum_{1 \leq i < j < k \leq n} P(X_{ijk} = 1) \end{aligned}$$

For a triangle $\{i, j, k\}$ to exist, all three edges must be present:

- Edge (i, j) exists with probability p
- Edge (i, k) exists with probability p
- Edge (j, k) exists with probability p
- By independence: $P(X_{ijk} = 1) = p \cdot p \cdot p = p^3$

Therefore:

$$E[T] = \binom{n}{3} p^3 = \frac{n(n-1)(n-2)}{6} p^3$$

Finding the Variance

Now we need to compute $\text{Var}(T)$.

Key formula for variance of a sum:

$$\text{Var} \left(\sum_i X_i \right) = \sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

Unlike expectation (where sums always split nicely), variance requires us to consider **covariances** between pairs of indicator variables.

Step 1: Compute $\text{Var}(X_{ijk})$ for a single triangle

For an indicator variable X with $P(X = 1) = p$:

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= E[X] - (E[X])^2 && \text{(since } X^2 = X \text{ for indicators)} \\ &= p - p^2 \\ &= p(1 - p) \end{aligned}$$

For our triangle indicators with $P(X_{ijk} = 1) = p^3$:

$$\text{Var}(X_{ijk}) = p^3(1 - p^3)$$

There are $\binom{n}{3}$ such triangles, so:

$$\sum_{1 \leq i < j < k \leq n} \text{Var}(X_{ijk}) = \binom{n}{3} p^3(1 - p^3)$$

Step 2: Compute covariances $\text{Cov}(X_{ijk}, X_{lmr})$

For two different triangles, we need:

$$\text{Cov}(X_{ijk}, X_{lmr}) = E[X_{ijk}X_{lmr}] - E[X_{ijk}]E[X_{lmr}]$$

Case 1: Triangles share 0 or 1 vertices

If triangles $\{i, j, k\}$ and $\{\ell, m, r\}$ share 0 or 1 vertices, they have **no edges in common**. Since all edges are independent:

$$\begin{aligned} E[X_{ijk}X_{lmr}] &= P(\text{both triangles exist}) \\ &= P(\text{triangle } \{i, j, k\} \text{ exists}) \cdot P(\text{triangle } \{\ell, m, r\} \text{ exists}) \\ &= p^3 \cdot p^3 \\ &= E[X_{ijk}] \cdot E[X_{lmr}] \end{aligned}$$

Therefore: $\text{Cov}(X_{ijk}, X_{lmr}) = 0$ (they are independent).

Case 2: Triangles share exactly 2 vertices (one edge)

If triangles $\{i, j, k\}$ and $\{i, j, \ell\}$ share the edge (i, j) , they are **NOT independent**.

For both triangles to exist, we need:

- Edge (i, j) exists (shared by both)
- Edge (i, k) exists
- Edge (j, k) exists
- Edge (i, ℓ) exists
- Edge (j, ℓ) exists

This is a total of 5 edges (not 6, because (i, j) is shared). Therefore:

$$E[X_{ijk}X_{ij\ell}] = p^5$$

The covariance is:

$$\begin{aligned} \text{Cov}(X_{ijk}, X_{ij\ell}) &= E[X_{ijk}X_{ij\ell}] - E[X_{ijk}]E[X_{ij\ell}] \\ &= p^5 - p^3 \cdot p^3 \\ &= p^5 - p^6 \\ &= p^5(1 - p) \end{aligned}$$

How many such pairs are there?

For each edge (i, j) in the graph:

- There are $\binom{n}{2}$ possible edges
- For a given edge (i, j) , we need to count pairs of triangles that both contain this edge
- Each triangle containing edge (i, j) is formed by adding a third vertex $k \neq i, j$
- There are $n - 2$ choices for such a third vertex
- The number of ways to choose 2 such vertices is $\binom{n-2}{2} = \frac{(n-2)(n-3)}{2}$

Total number of pairs sharing exactly one edge:

$$\binom{n}{2} \cdot \binom{n-2}{2} = \frac{n(n-1)}{2} \cdot \frac{(n-2)(n-3)}{2} = \frac{n(n-1)(n-2)(n-3)}{4}$$

Step 3: Combine everything

$$\begin{aligned} \text{Var}(T) &= \sum \text{Var}(X_{ijk}) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \\ &= \binom{n}{3} p^3 (1 - p^3) + \frac{n(n-1)(n-2)(n-3)}{4} p^5 (1 - p) \end{aligned}$$

Simplifying:

$$\text{Var}(T) = \binom{n}{3} p^3 (1 - p^3) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p)$$

Or equivalently:

$$\text{Var}(T) = \frac{n(n-1)(n-2)}{6} p^3 (1 - p^3) + \frac{n(n-1)(n-2)(n-3)}{4} p^5 (1 - p)$$

Key Insights

1. The variance has **two terms**:
 - First term: variance of individual indicators
 - Second term: covariances between dependent pairs
2. **Why covariances matter**: Triangles that share an edge are positively correlated. If edge (i, j) exists, it increases the probability that both triangles containing it exist.
3. **Pairs that share 0 or 1 vertices don't contribute** because they are independent (covariance = 0).
4. The problem demonstrates that **dependence structure matters** for variance, even though it doesn't affect expectation (due to linearity).

Alternative Approach: Using $E[T^2]$ (Easier!)

Instead of computing all the covariances explicitly, we can use the simpler formula:

$$\text{Var}(T) = E[T^2] - (E[T])^2$$

This approach is more direct because we just need to compute $E[T^2]$, which counts **pairs of triangles**.

Step 1: Recall $E[T]$ and compute $(E[T])^2$

From the previous homework:

$$E[T] = \binom{n}{3} p^3$$

Therefore:

$$(E[T])^2 = \binom{n}{3}^2 p^6$$

Step 2: Compute $E[T^2]$

Recall that $T = \sum_{\{i,j,k\}} X_{ijk}$ where X_{ijk} are indicator variables for each triangle.

$$T^2 = \left(\sum_{\{i,j,k\}} X_{ijk} \right)^2 = \sum_{\{i,j,k\}} X_{ijk}^2 + \sum_{\{i,j,k\} \neq \{\ell,m,r\}} X_{ijk} X_{\ell mr}$$

Taking expectations:

$$E[T^2] = \sum_{\{i,j,k\}} E[X_{ijk}^2] + \sum_{\{i,j,k\} \neq \{\ell,m,r\}} E[X_{ijk} X_{\ell mr}]$$

First term: $\sum E[X_{ijk}^2]$

Since X_{ijk} is an indicator, $X_{ijk}^2 = X_{ijk}$, so:

$$E[X_{ijk}^2] = E[X_{ijk}] = p^3$$

There are $\binom{n}{3}$ triangles, so:

$$\sum_{\{i,j,k\}} E[X_{ijk}^2] = \binom{n}{3} p^3$$

Second term: $\sum E[X_{ijk} X_{\ell mr}]$ for different triangles

We need $E[X_{ijk} X_{\ell mr}] = P(\text{both triangles exist})$.

This depends on how many vertices the two triangles share:

Case A: Triangles share 0 or 1 vertices (no common edges)

If they have no edges in common, the events are independent:

$$E[X_{ijk} X_{\ell mr}] = p^3 \cdot p^3 = p^6$$

Case B: Triangles share exactly 2 vertices (share 1 edge)

If triangles $\{i, j, k\}$ and $\{i, j, \ell\}$ share edge (i, j) , then for both to exist we need 5 edges total:

$$E[X_{ijk}X_{ij\ell}] = p^5$$

How many pairs of each type?

Total pairs of different triangles:

$$\binom{\binom{n}{3}}{2} = \frac{1}{2} \binom{n}{3} \left(\binom{n}{3} - 1 \right)$$

Pairs sharing exactly one edge (Case B):

$$\binom{n}{2} \cdot \binom{n-2}{2}$$

Pairs sharing 0 or 1 vertices (Case A):

$$\binom{\binom{n}{3}}{2} - \binom{n}{2} \binom{n-2}{2}$$

Step 3: Combine to get $E[T^2]$

$$\begin{aligned} E[T^2] &= \binom{n}{3} p^3 + \left[\binom{\binom{n}{3}}{2} - \binom{n}{2} \binom{n-2}{2} \right] p^6 + \binom{n}{2} \binom{n-2}{2} p^5 \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 + \binom{n}{2} \binom{n-2}{2} (p^5 - p^6) \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \end{aligned}$$

Step 4: Compute the variance

Using $\binom{\binom{n}{3}}{2} = \frac{1}{2} \binom{n}{3} (\binom{n}{3} - 1) = \frac{1}{2} \binom{n}{3}^2 - \frac{1}{2} \binom{n}{3}$:

$$\begin{aligned} \text{Var}(T) &= E[T^2] - (E[T])^2 \\ &= \binom{n}{3} p^3 + \left[\frac{1}{2} \binom{n}{3}^2 - \frac{1}{2} \binom{n}{3} \right] p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) - \binom{n}{3}^2 p^6 \\ &= \binom{n}{3} p^3 + \frac{1}{2} \binom{n}{3}^2 p^6 - \frac{1}{2} \binom{n}{3} p^6 - \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 - \frac{1}{2} \binom{n}{3} p^6 - \frac{1}{2} \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 \left(1 - \frac{1}{2} p^3 \right) - \frac{1}{2} \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \end{aligned}$$

Simplifying further:

$$\text{Var}(T) = \binom{n}{3} p^3 (1 - p^3) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p)$$

$$\text{Var}(T) = \binom{n}{3} p^3 (1 - p^3) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p)$$

This matches our previous answer!

Why This Approach is Easier

1. **More direct:** We compute $E[T^2]$ by counting pairs of triangles, rather than tracking individual covariances
2. **Cleaner bookkeeping:** Only need to classify pairs by overlap (0-1 vertices vs 2 vertices)
3. **Single formula:** Everything flows from $\text{Var}(T) = E[T^2] - (E[T])^2$
4. **Same result:** Both methods give the same answer, but this one feels less tedious
5. **Better intuition:** Counting pairs of outcomes is often more natural than computing covariances explicitly

Key insight: When dealing with sums of indicators, computing $E[T^2]$ means counting **ordered pairs of outcomes**. For indicators, $E[X_i X_j] = P(\text{both occur})$. The trick is to **count by cases** based on overlap structure. This is essentially the same as the covariance approach, but with less explicit covariance notation!